Auteur: Ahmed Meiyaki
Studierichting: Informatica
Oefeningenreeks 3G nr. 11.6

$$f(x) = \frac{x^2}{x-9}$$

$\Rightarrow$ VA: $x = 9$

$$\lim_{x \to 9+} f(x) = +\infty$$

$$\lim_{x \to 9-} f(x) = -\infty$$

$\Rightarrow$ HA: geen

$$SA: m = \lim_{x \to \infty} \frac{f(x)}{x} = \lim_{x \to \infty} \frac{x^2}{x^2 - 9x} = 1$$

$$q = \lim_{x \to \infty} (f(x) - x) = \lim_{x \to \infty} \frac{x^2}{x-9} - x$$

$$= \lim_{x \to \infty} \frac{x^2}{x-9} - \frac{x^2 - 9x}{x-9} = \lim_{x \to \infty} \frac{9x}{x-9} = 9$$

$\Rightarrow$ SA: $y = x + 9$

Ligging:

x		9	
	$-$	/	$+$

$x \to +\infty \Rightarrow f(x) > SA$

$x \to -\infty \Rightarrow f(x) < SA$

$$f'(x) = \frac{2x(x-9) - x^2}{(x-9)^2} = \frac{2x^2 - 18x - x^2}{x^2 - 18x + 81}$$

$$= \frac{x^2 - 18x}{(x-9)^2}$$

$$f''(x) = \frac{(2x-18)\left(x^2 - 18x + 81\right) - \left(x^2 - 18x\right)(2x - 18)}{(x-9)^4}$$

$$= \frac{(2x-18)\left(x^2 - 18x + 81 - x^2 + 18x\right)}{(x-9)^4}$$

$$= \frac{2(x-9) \cdot 81}{(x-9)^3} = \frac{162}{(x-9)^2}$$

x		0		9		18	
$f(x)$	$-$	0	$-$	|	$+$	$+$	$+$
$f'(x)$	$+$	0	$-$	|(2)	$-$	0	$+$
$f''(x)$	$-$	$-$	$-$	|(3)	$+$	$+$	$+$
	$\nearrow$	$M(0)$ $\frown$	$\searrow$	|	$\searrow$	$m(36)$ $\smile$	$\searrow$

Grafiek:

3G-11.6-ahmed.meiyaki.INF.png

References